Exp 9: Implement a classification model to distinguish between normal and abnormal physiological signals using extracted signal features. (perform comparative analysis on different ML models)

Dataset Link: https://www.kaggle.com/datasets/shayanfazeli/heartbeat

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix,classification_report,roc
import pickle
```

```
df=pd.read_csv('/content/drive/MyDrive/mitbih_test.csv',header=None)
df.columns=[f'x{i}' for i in range(187)]+['y']
print(df.head())
```

```
         x0        x1        x2        x3        x4        x5        x6  \
0  1.000000  0.758264  0.111570  0.000000  0.080579  0.078512  0.066116
1  0.908425  0.783883  0.531136  0.362637  0.366300  0.344322  0.333333
2  0.730088  0.212389  0.000000  0.119469  0.101770  0.101770  0.110619
3  1.000000  0.910417  0.681250  0.472917  0.229167  0.068750  0.000000
4  0.570470  0.399329  0.238255  0.147651  0.000000  0.003356  0.040268

         x7        x8        x9  ...  x178  x179  x180  x181  x182  x183  \
0  0.049587  0.047521  0.035124  ...   0.0   0.0   0.0   0.0   0.0   0.0
1  0.307692  0.296703  0.300366  ...   0.0   0.0   0.0   0.0   0.0   0.0
2  0.123894  0.115044  0.132743  ...   0.0   0.0   0.0   0.0   0.0   0.0
3  0.004167  0.014583  0.054167  ...   0.0   0.0   0.0   0.0   0.0   0.0
4  0.080537  0.070470  0.090604  ...   0.0   0.0   0.0   0.0   0.0   0.0

   x184  x185  x186    y
0   0.0   0.0   0.0  0.0
1   0.0   0.0   0.0  0.0
2   0.0   0.0   0.0  0.0
3   0.0   0.0   0.0  0.0
4   0.0   0.0   0.0  0.0

[5 rows x 188 columns]
```

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21892 entries, 0 to 21891
Columns: 188 entries, x0 to y
dtypes: float64(188)
memory usage: 31.4 MB
```

	x0	x1	x2	x3	x4	x5	x6	x7	
count	21892.000000	21892.000000	21892.000000	21892.000000	21892.000000	21892.000000	21892.000000	21892.000000	21892.000
mean	0.894410	0.761902	0.426627	0.221596	0.201676	0.209891	0.204805	0.200992	0.197
std	0.234560	0.218659	0.228572	0.208711	0.177727	0.172194	0.177946	0.176142	0.170
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000
25%	0.924260	0.683366	0.251197	0.050505	0.082873	0.087912	0.072663	0.065997	0.064
50%	0.990431	0.828996	0.432777	0.167630	0.147642	0.158111	0.144068	0.144509	0.150
75%	1.000000	0.912319	0.583991	0.347092	0.259211	0.287356	0.298453	0.294563	0.289
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000

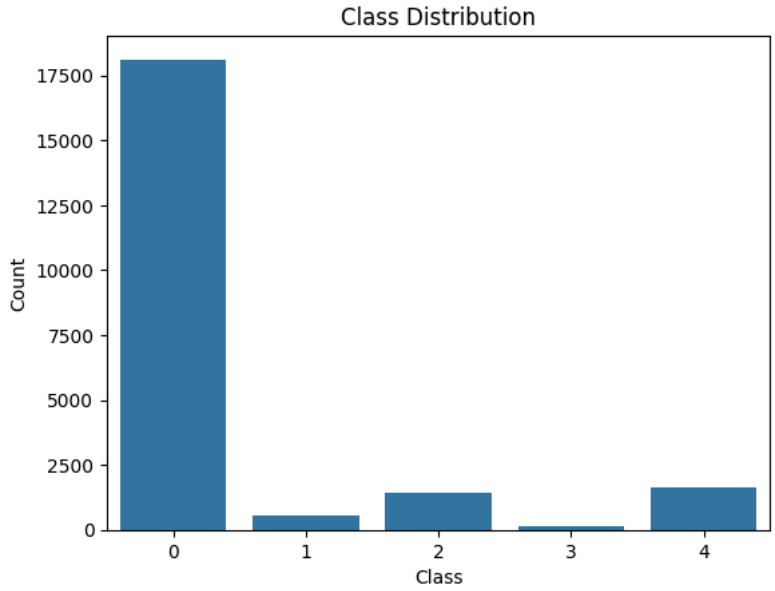
8 rows × 188 columns

```
df['y']=df['y'].astype(int)
print(df['y'].value_counts())
```

```
y
0    18118
4     1608
2     1448
```

```
1      556
3      162
Name: count, dtype: int64
```

```
sns.countplot(x='y',data=df)
plt.title("Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")
plt.show()
```




```
feature_cols=[f'x{i}' for i in range(187)]
```

```
print(df[feature_cols].isnull().sum().sum())
```

```
0
```

```
X=df[feature_cols]
y=df['y']
```

```
sc=StandardScaler()
X_scaled=sc.fit_transform(X)
```

```
X_train,X_test,y_train,y_test=train_test_split(X_scaled,y,test_size=0.2,random_state=0,stratify=y)
print("Train size:",X_train.shape)
print("Test size:",X_test.shape)
```

```
Train size: (17513, 187)
Test size: (4379, 187)
```

```
models={
"Logistic Regression":LogisticRegression(max_iter=1000),
"Random Forest":RandomForestClassifier(),
"SVM":SVC(),
"KNN":KNeighborsClassifier(),
"XGBoost":XGBClassifier(eval_metric='mlogloss')
}
```

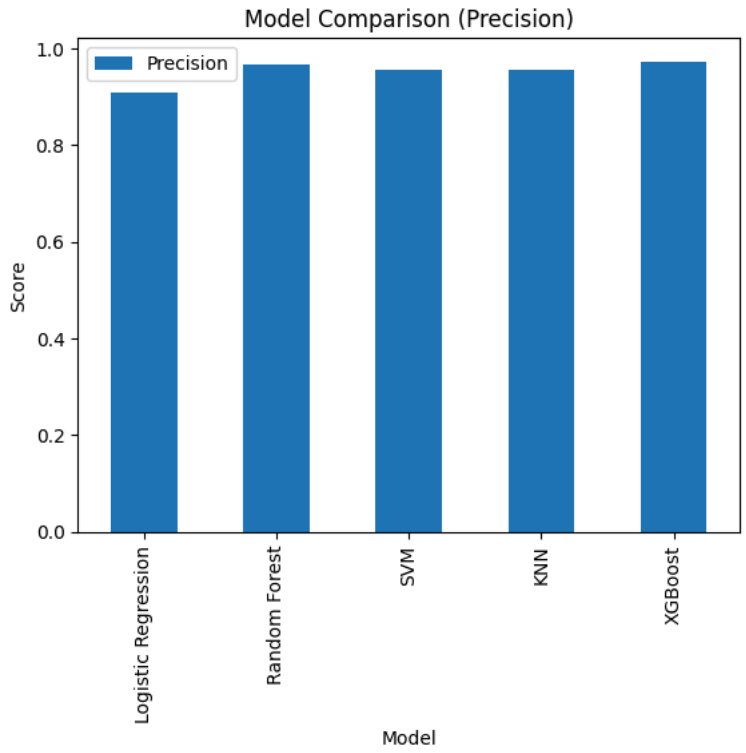
```
results=[]
for name,model in models.items():
    model.fit(X_train,y_train)
    pred=model.predict(X_test)
    acc=accuracy_score(y_test,pred)
    pre=precision_score(y_test,pred,average='weighted')
    rec=recall_score(y_test,pred,average='weighted')
    f1w=f1_score(y_test,pred,average='weighted')
    f1m=f1_score(y_test,pred,average='macro')
    f1mi=f1_score(y_test,pred,average='micro')
    results.append([name,acc,pre,rec,f1w,f1m,f1mi])
```

```
results_df=pd.DataFrame(results,columns=["Model","Accuracy","Precision","Recall","F1_weighted","F1_macro","F1_micro"])
results_df
```

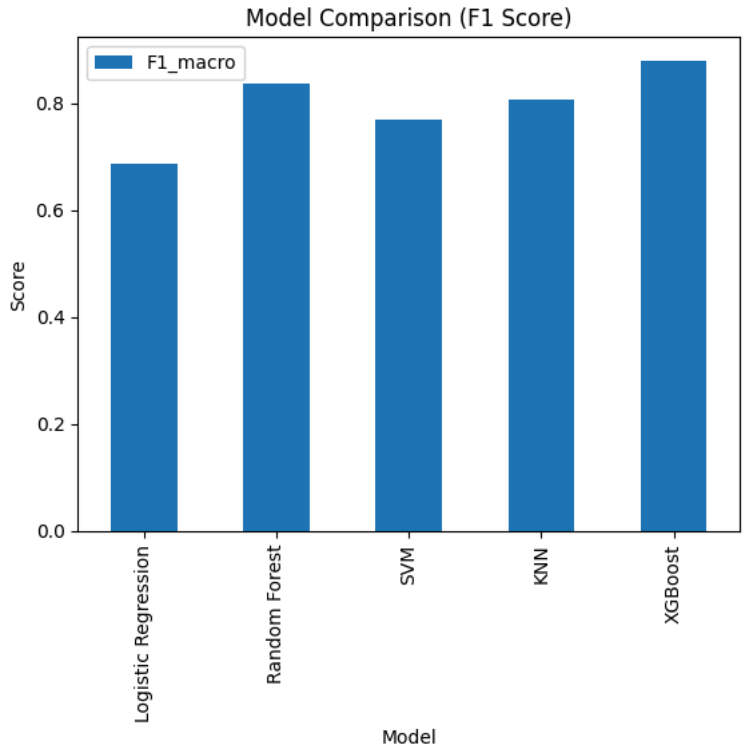
	Model	Accuracy	Precision	Recall	F1_weighted	F1_macro	F1_micro
0	Logistic Regression	0.917561	0.908050	0.917561	0.907757	0.686894	0.917561
1	Random Forest	0.967116	0.967232	0.967116	0.964763	0.836542	0.967116
2	SVM	0.955926	0.955235	0.955926	0.952063	0.769989	0.955926
3	KNN	0.958210	0.956327	0.958210	0.956085	0.806425	0.958210
4	XGBoost	0.973282	0.973215	0.973282	0.971709	0.880158	0.973282

Next steps: Generate code with results_df New interactive sheet

```
results_df.plot(x="Model",y="Precision",kind="bar")
plt.title("Model Comparison (Precision)")
plt.ylabel("Score")
plt.show()
```




```
results_df.plot(x="Model",y="F1_macro",kind="bar")
plt.title("Model Comparison (F1 Score)")
plt.ylabel("Score")
plt.show()
```

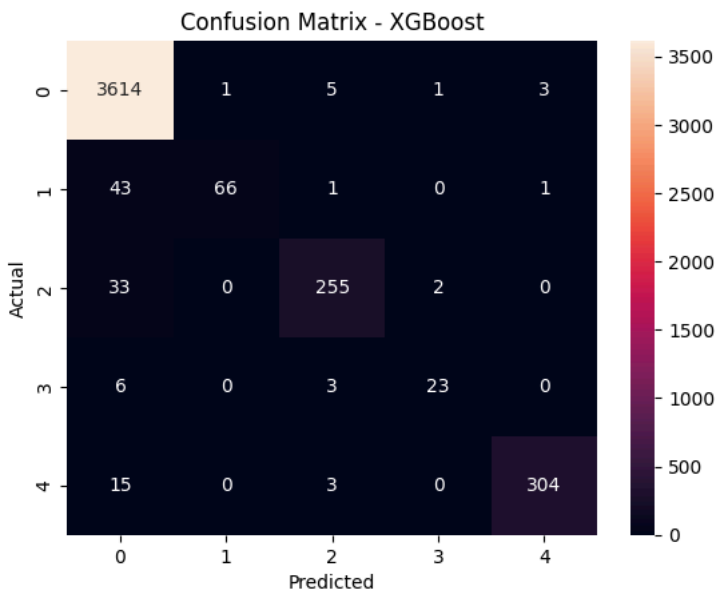



```
best_name=results_df.sort_values("F1_macro",ascending=False).iloc[0,0]
best_model=models[best_name]
print("Best Model:",best_name)
```

```
Best Model: XGBoost
```

```
with open("heartbeat_classifier.pkl","wb") as f:
    pickle.dump(best_model,f)
```

```
pred=best_model.predict(X_test)
cm=confusion_matrix(y_test,pred)
sns.heatmap(cm,annot=True,fmt='d')
plt.title("Confusion Matrix - "+best_name)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```




```
print(classification_report(y_test,pred))
```

```
              precision    recall  f1-score   support

           0       0.97      1.00      0.99      3624
           1       0.99      0.59      0.74       111
           2       0.96      0.88      0.92       290
```

```
                3       0.88      0.72      0.79        32
```